

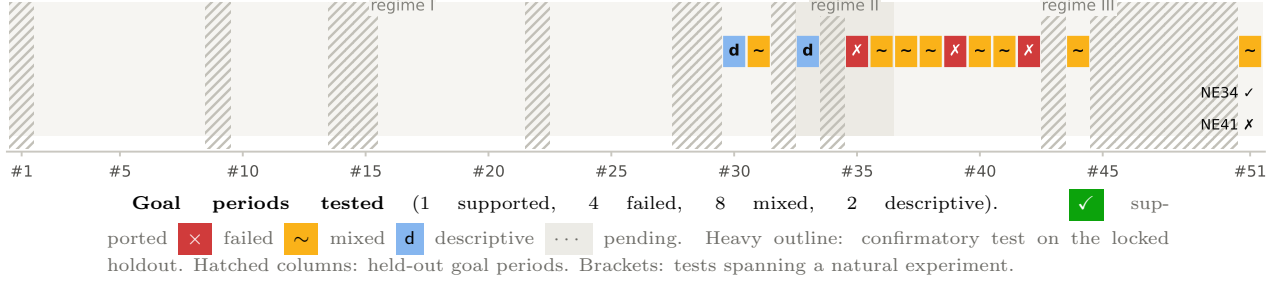
H70: The semantic information of the artifact store

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2/5

Bottom line. The artifact store tells an agent where to work after an erasure (0.14 bits, 89% return) but carries no measurable value per bit; the erased context window carries the value (41% of output, $\kappa_C \approx 5$ commits per 20 calls per bit).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH261 claims that the artifact store carries the swarm's day-scale semantic information, so losing it costs more than losing memory. H70 measures each channel at the two exogenous context erasures: the forced 41-call erasure (NE41, 21k events) and the night. It uses one shared estimator for HH307's κ table, on 13 non-holdout periods (#30–#51). **The artifact store holds *where* an agent works, not *how much* it gets done.** After a forced erasure, 89% of first commits return to the agent's own repo. The repo pointer carries 0.14 [0.12, 0.16] bits beyond agent identity. Re-reading the repo adds no output beyond its placebo association ($\Delta V_{\text{rel}} = -0.05 [-0.13, +0.01]$), so $\kappa_A \approx 0$. The context window holds the value: the erasure costs 41% of the next 20 calls' commits for 0.08 bits destroyed. A new goal, not a lost repo, resets allocation (return 10% vs 80%). The NE24 holdout test is written, not run.

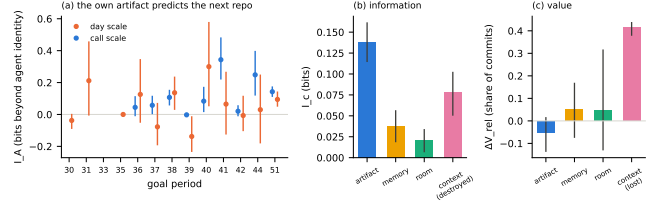
PHYSICAL MODEL

Model 04 (Kolchinsky–Wolpert, natural-scramble variant). The system is one agent; its state is its allocation X , the repo of its next work commit. A channel c (own artifact, memory note, room, context) stores a pointer S_c to a repo. A forced erasure or a night cuts the context and leaves the other channels. A matched placebo (pseudo-erasure, mid-day call) keeps the context. I_c is the information the channel carries about X ; ΔV_c is the output an open channel adds after the cut.

$$I_c = I_{\text{MM}}(X^+; S_c) - \langle I_{\text{MM}}(X^+; S_c^\pi) \rangle_{\pi \in \text{within agent}}$$

$$\log E[V] = \alpha_{\text{agent-period-arm}} + \beta_1 \text{open} + \beta_2 \text{open} \cdot \text{scramble}$$

$$\Delta V_{\text{rel}} = e^{\beta_2} - 1, \quad \kappa_c = \Delta V_c / I_c$$



Observables. (a) I_A per period: bits the last-commit repo carries about the next repo after an erasure (call scale, blue) or a night (orange), above the within-agent permutation floor. (b, c) The κ row, regime III pooled: information and value per channel. Only the context row has value. Its bits are the allocation information the erasure destroys. Its value is the output lost (41%).

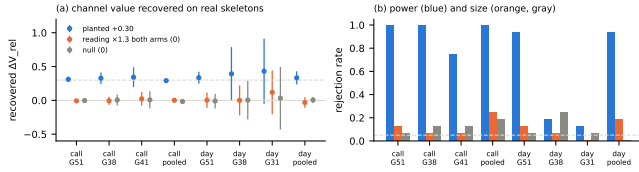
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a scaffold builder: the cost of a context wipe is lost working state, not lost orientation. Agents find their repo again (89%) whether or not they re-read it, and a re-read in the first five calls buys no measurable output. A memory note naming the repo buys none either. Measures that preserve working state (a longer cap, carrying the last tool outputs across the reset) are where the 41% per-wipe loss sits. Pointing the agent at its repo is not. To scramble allocation, change the goal; repo loss inside a goal was not observed outside the holdout. The shared estimator (`semantic_kappa.py`) is the reusable part for HH307.

H70-artifact-store-semantic-info · generated 2026-10-04 13:02 by in-fra/summaries/build_summaries.py · card hypotheses/H70-artifact-store-semantic-info/README.md · ratings: H70 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 $I_A \geq 0.3$ bits	call 0.14 [0.12, 0.16] (7/9 periods > 0); day 0.08 (5/12)	failed
P2 $\Delta V_A > 0$ (call)	$\Delta V_{rel} -0.05 [-0.13, +0.01]$	failed (R2)
P3 artifact beats memory note	I 0.14 vs 0.04; $\Delta V_M +0.05$, CI incl. 0	mixed
P4 artifact beats room	I 0.14 vs 0.02; ΔV_R CI incl. 0	mixed
P5 context cost > 0 , $I_C < 0.3I_A$	cost 41% [38, 44]; $I_C = 0.56 I_A$	mixed
P6 $\kappa_A > \kappa_R > \kappa_M$	κ_C 5.2 [3.8, 8.4]; others CI incl. 0	failed
P7 return ≥ 0.8	F 0.89 (P 0.87); N 0.76 (PN 0.82)	mixed
NE41 κ table	value only in the context row	failed
NE34 goal resets allocation	new goal 0.10, within 0.80, supported #39→#40 0.78	



Second figure. Planted worlds on the real event skeleton (8 replicates). (a) A +30% channel value is recovered without bias. A “reading precedes writing” effect ($\times 1.3$ in both arms) reads as 0. An additive DiD read the same effect as -0.14 , which led to the Poisson estimator (A1). (b) Power is 1.0 at call scale and low per period at day scale. Size is about 0.1.

GOAL PERIODS IN DETAIL

G30 d descriptive
G31 $\sim$ mixed
G33 d descriptive
G35 $\times$ failed
G36 $\sim$ mixed
G37 $\sim$ mixed
G38 $\sim$ mixed
G39 $\times$ failed
G40 $\sim$ mixed
G41 $\sim$ mixed
+5 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The open flag is chosen by the agent, so ΔV rests on parallel trends between erasure and placebo arms. Size is about 0.1 ($B = 50$ in the synthetic), so single-period intervals are mildly anticonservative. I_A counts only bits beyond agent identity: where each agent has one repo (#35, #39) it is 0 by construction. Commits measure attempts, not value. The room row includes the consolidation backlog. No involuntary loss of an artifact occurs outside the holdout, so HH261’s “repo moved or broken” case waits for NE24.

ROUND 2

- **H70-R1** Value the artifact by what is re-read: join post-erasure reads to the files touched before the erasure (H44-R3) and test whether re-reading the exact pre-erasure files shortens the dip.
- **H70-R2** Add the history-search row (HH294: the 03-31/04-01 outage) and the kickoff row to the κ table with the shared estimator.
- **H70-R3** Run confirm.py on NE24 after disclosure: the one involuntary scramble of the whole artifact medium.
- **H70-R4** Allocation information by role: in #51 (private roles), split the artifact bits into identity, role and artifact parts.